

(15 Points)

Q1-Circle the right answer then fill the table below

- 1-A fuse is never inserted in _____?
A. neutral wire B. negative of D.C. circuit C. positive of D.C. circuit D. phase line
- 2-A fuse is connected _____?
A. in series with circuit B. in parallel with circuit C. either in series or in parallel with circuit D. none of the above
- 3-The material used for fuse must have _____?
A. low melting point and high specific resistance B. low melting point and low specific resistance
C. high melting point and low specific resistance D. low melting point and any specific resistance
- 4-On which of the following effects of electric current a fuse operates?
A. Photoelectric effect B. Electrostatic effect C. Heating effect D. Magnetic effect
- 5-What is the main purpose of an electrical switch?
a) To convert AC to DC b) To control the flow of electricity
c) To convert DC to AC d) To measure the current flow
- 6-Which type of switch controls one or more lights from a single location?
a) Single-pole switch b) Double-pole switch c) Three-way switch d) Four-way switch
- 7-What type of switch allows you to control a light from two different locations?
a) Single-pole switch b) Double-pole switch c) Three-way switch d) Four-way switch
- 8-Which type of switch operates by physically moving a conductor to open or close a circuit?
a) Solid-state switch b) Mechanical switch c) Semiconductor switch d) Transistor switch
- 9-What type of switch is commonly used in test equipment because it can connect one input to multiple outputs?
a) Single-pole switch b) Double-pole switch c) Rotary switch d) Dimmer switch
- 10-In a circuit with multiple switches connected in parallel, how many switches need to be closed to complete the circuit?
a) All of them b) Any one of them c) At least half of them d) Two-thirds of them
- 11-Function without physical or mechanical contact?
a) Solid-state switch b) Mechanical switch c) Rotary switch d) Toggle switch
- 12-What is a mercury switch?
a) A switch that uses mercury as a dielectric medium
b) A switch that uses mercury to open or close a circuit
c) A switch sensitive to mercury level
d) A switch that operates under high temperatures
- 13-What type of switch is commonly used in home automation systems?
a) Toggle switch b) Push-button switch c) Relay switch d) Smart switch
- 14-Which switch allows the current to flow in only one direction?
a) Diode b) Transistor c) Resistor d) Capacitor

15. Solid state relays have only one moving part which helps in switching between "ON" and "OFF" position.
a) True b) False

16. A good electrical relay contact tip should have higher arc resistance.
a) True b) False

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	A	X	X	B	D	X	B	C	B	A	B	A	A	A

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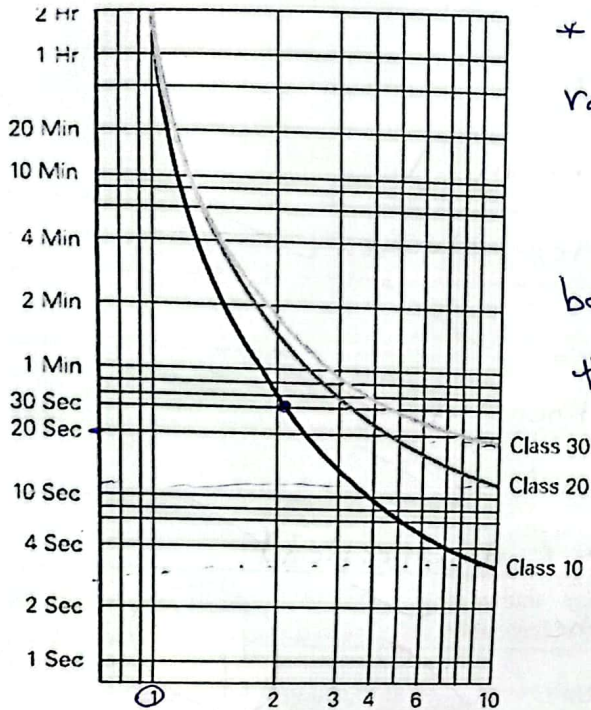
Q2: What are the meaning of the commonly used abbreviations (5Points)

1	THR	thermal Relay ✓
2	TR	thermal Relay X
3	MSP	Motor starter protection ✓
4	GEN	Generator ✓
5	RH	Relay Rheostat ✓
6	TB	Terminal Board ✓
7	DISC	Disconnected (disconnect) ✓
8	FTS	foat switch ✓
9	SOL	solenoid ✓
10	PS	Pressure switch ✓

Q3-Name the symbols below using the upper square(5Points)

disconnected switch	tripportals switch	circuit breaker with Thermal overload	circuit breaker with the thermal and magnetic overload	circuit breaker with the thermal and magnetic overload

Q4-Using the figure below explain the rated trip class of overload relay(5Points)



* the over load relay have
rated trip classes
 → class 10
 → class 20
 → class 30

base on rated current and
the time that take to tripp

كلما زاد التيار في class 30

تدريسي trip كان 20sec

في class 20 كان trip 11sec

في class 10 كان trip 3sec

كلما زاد التيار في class 30 الوقت اللازم لحدوث trip

كلما زاد التيار في class 20 الوقت اللازم لحدوث trip

كلما زاد التيار في class 10 الوقت اللازم لحدوث trip

logarithmic function